Roofline Model
How It Helps in Optimizations

- Courtesy for some slides: S. Williams, Xiaoye Sherry Li and others

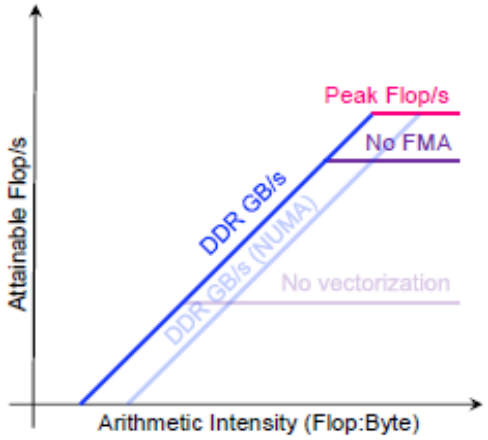
Slides Adapted By: Sparsh Mittal

Acronyms

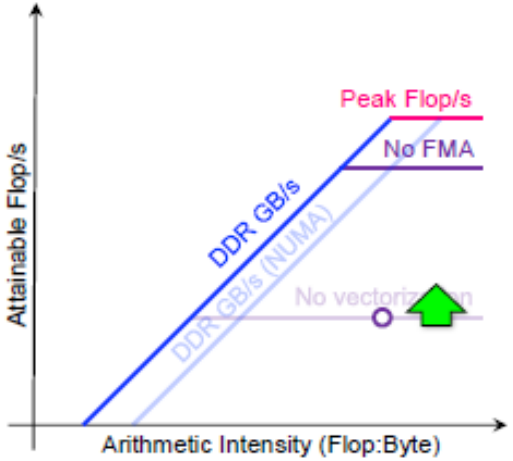
- AI – Arithmetic intensity
- DP – Double precision
- SIMD – Single-instruction multiple data
- ILP – Instruction-level parallelism
- FMA – Fused multiply add
- FP – Floating-point
- NUMA – Non-uniform memory access
- VPU – Vector processing unit

Three Broad Categories of Optimization Using Roofline Model

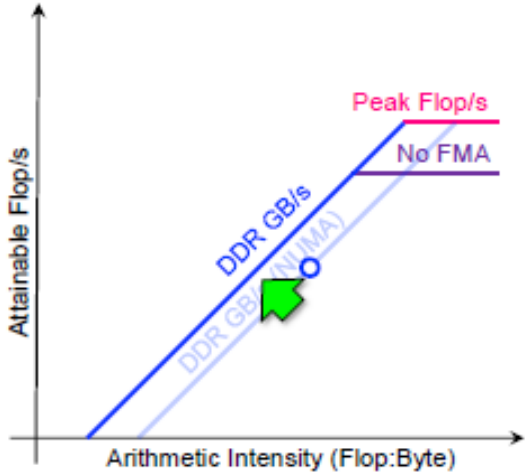
Three categories of optimizations



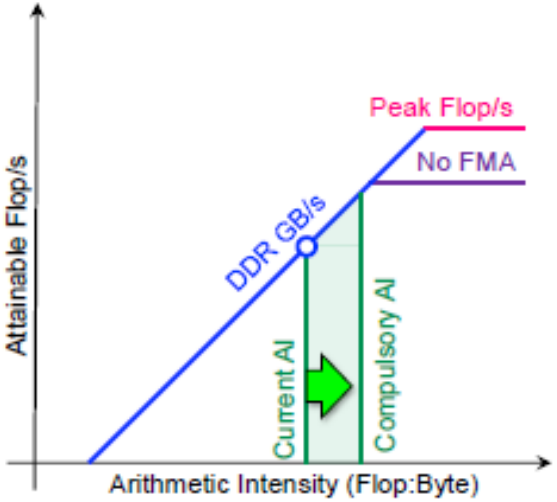
Original roofline model



Boost in-core performance (e.g., vectorize using compiler)

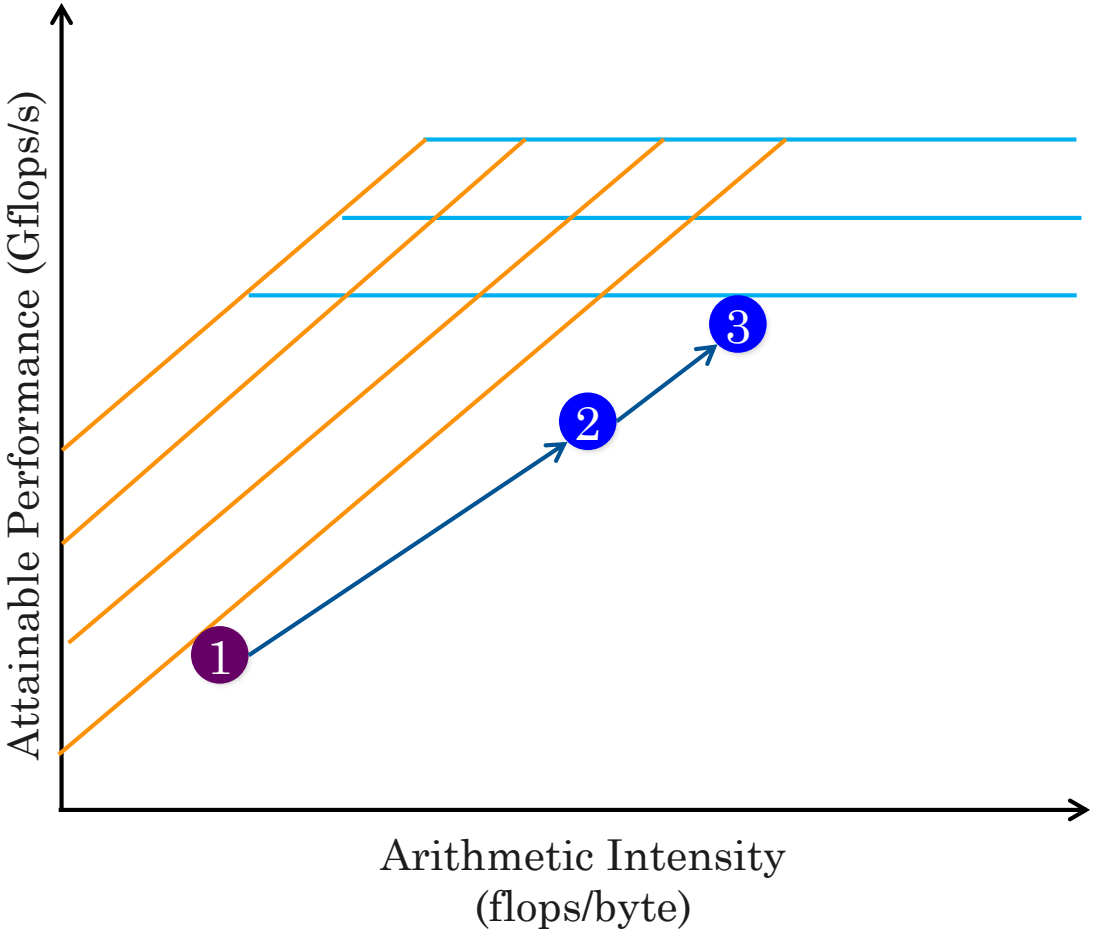


Boost memory bandwidth (e.g., NUMA-aware policies)



Reduce data-fetched from memory (e.g., by cache blocking)

Example 1: Memory Bound Application

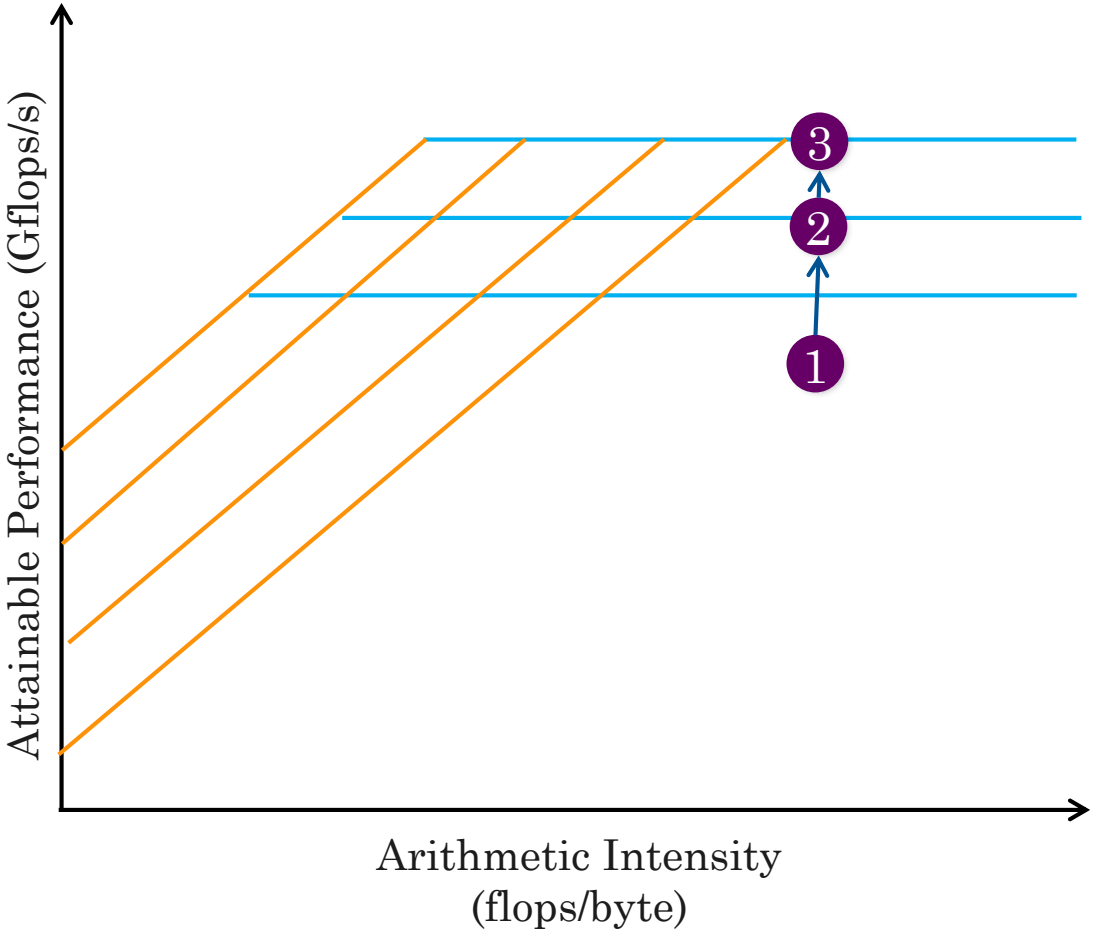


1. Low AI application. No cache reuse
   → DRAM bandwidth bound

2. Implement L2 cache optimization
   - L2 Cache is fully reused, performance increases
     → roofline moves to the right because we are doing less loads from DRAM.
3. Implement L1 cache optimization
   - We now hit computation ceiling

[1] S. Williams et al. *CACM* (2009), crd.lbl.gov/departments/computer-science/PAR/research/roofline

Example 2: Compute Bound Application



1. High AI application.
   - Compute bound but not using vectorization/FMA/both VPUs
2. Implement vectorization
   - Since we are not touching memory, the AI does not change
   - We are fully utilizing VPUs → Performance increases
3. Use FMAs

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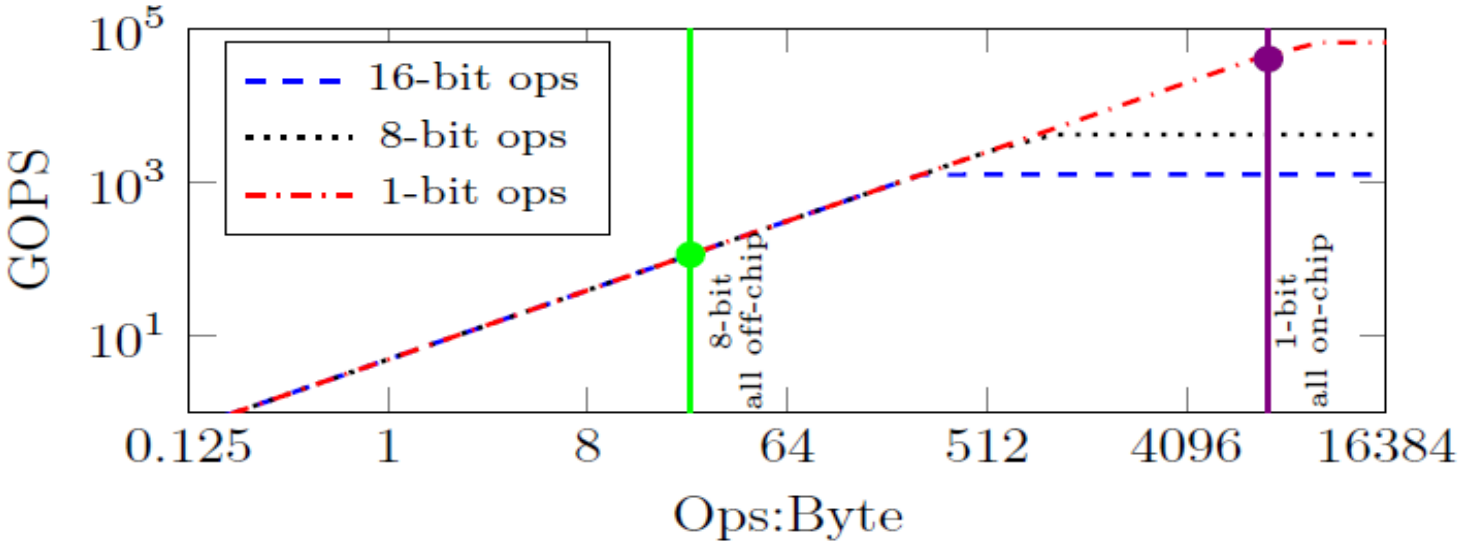
Impact of Lowering the Precision

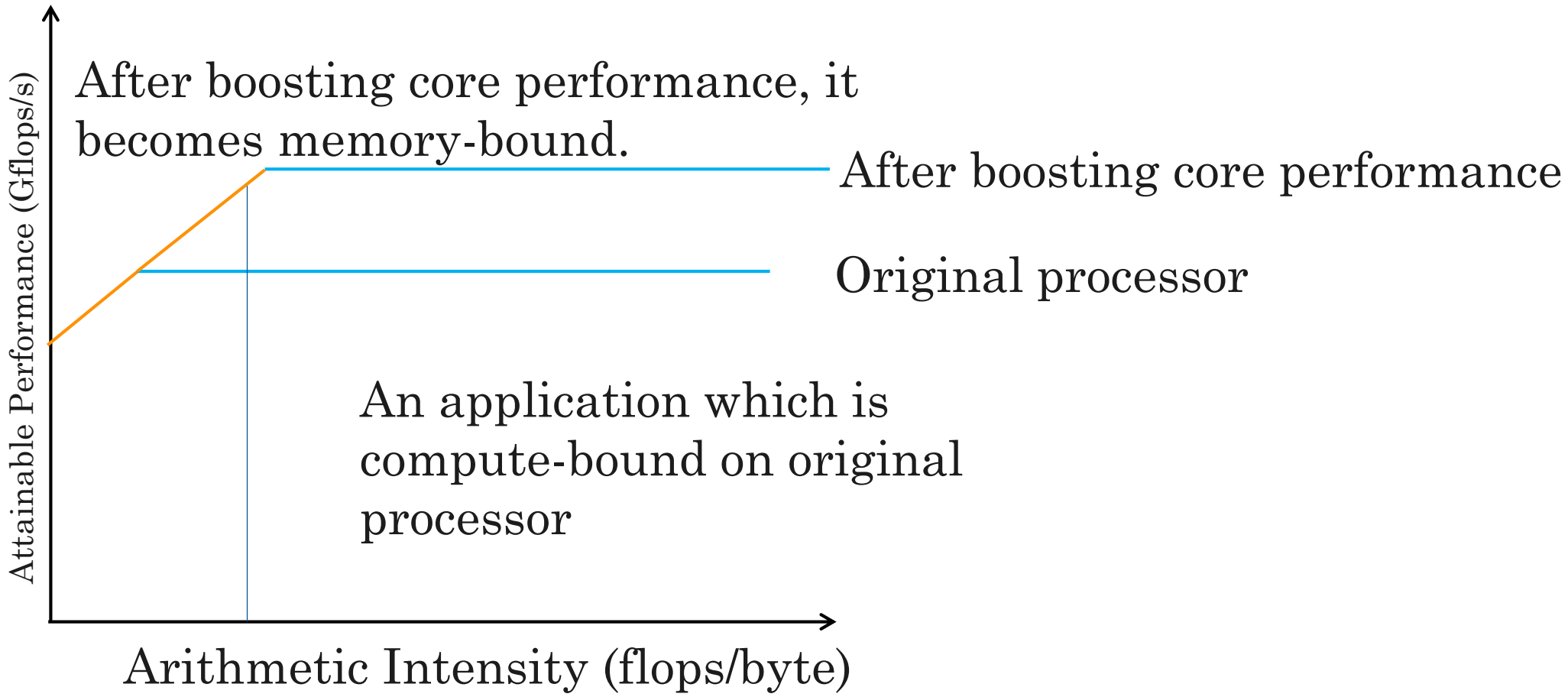
- For AlexNet, compute-bound performance for binary (1b) operations is
  - ➢16 times higher than 8b fixed-point and
  - ➢53 times higher than 16b fixed-point operations.



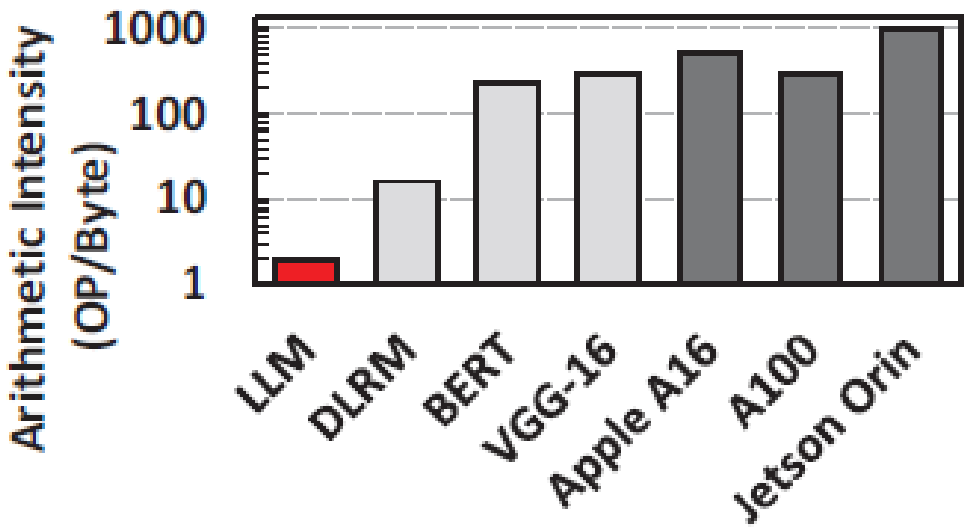
Roofline model of an FPGA. Vertical lines of AI are for AlexNet

An Example on How Roofline Model Helps Us in Designing a Balanced Processor

Achieving Balance Between Compute and Memory



Lesson: optimizing only one of memory or compute is not good. We need to optimize both and balance them.



Arithmetic Intensity comparison between LLM and other AI algorithms